

MBT55: From Microorganisms to Civilliation's OS

Redesigning Planetary-Scale Life Support Processes

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A Scene of Despair: Farmland After the 2011 Great East Japan Earthquake

- The tsunami from the Great East Japan Earthquake brought catastrophic damage to coastal farmlands. Farmland in Iwanuma City was covered in a thick 4-10cm layer of seabed sludge.
- This sludge contained high concentrations of salt, heavy metals, and petroleum compounds, severely contaminating the soil and making crop production considered impossible.
- This was not merely infrastructure damage; it signified the collapse of the very foundations of local community life.



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Recorded Reality: Sludge Disappears, Rice Planting Possible in 3 Weeks

- The microbial humus compost 'MBT55' was applied to this desperate soil.
- In just one week after application, the surface sludge layer rapidly decomposed and disappeared.
- By the 21st day after application (May 20), rice planting was carried out in the field that was once covered in sludge.
- By the 50th day (June 18), while the untreated area remained barren, rice was growing vigorously in the MBT55 treated area. This is the result of MBT55 initiating an ecological hypercycle that decomposes organic motter in the sludge and detoxifies harmful substances.



Untreated Area



MBT55 Treated Area

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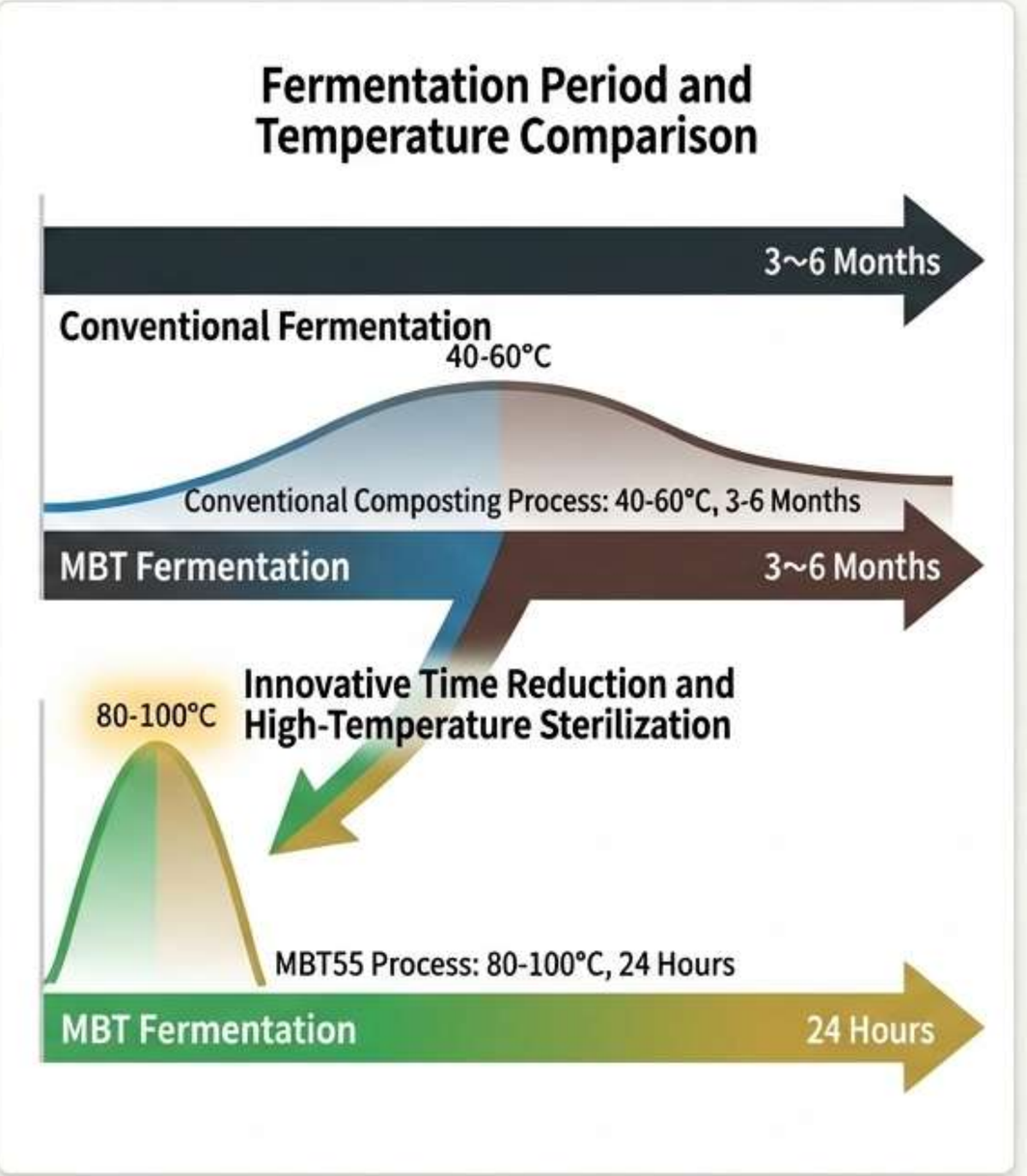
The Engine of Regeneration: The Science of MBT55

MBT55 is not just compost. It is a precisely engineered "Ecological Hypercycle Engine."

Proprietary Microbial Consortium: Consists of over 120 species of microorganisms composed in a ratio of 55% aerobic and 45% anaerobic. This balance enables simultaneous decomposition and stabilization.

24-Hour High-Temperature Fermentation: Powered by proprietary patented thermophilic and halophilic bacteria (e.g., Tateyama Ken, Tateyama Ryuou) that are active in high-temperature environments of 80-100°C. This completes the process that normally takes 3-6 months for composting in just 24 hours, completely inactivating pathogens and weed seeds.

M³-BioSynergy System: This engine functions as a system where three layers—① **Microbial Symbiotic Network** (Microbial), ② Nutrient Cascade (Metabolic), and ③ **Self-Circulating Dynamic Equilibrium** (Modular)—generate synergistic effects.

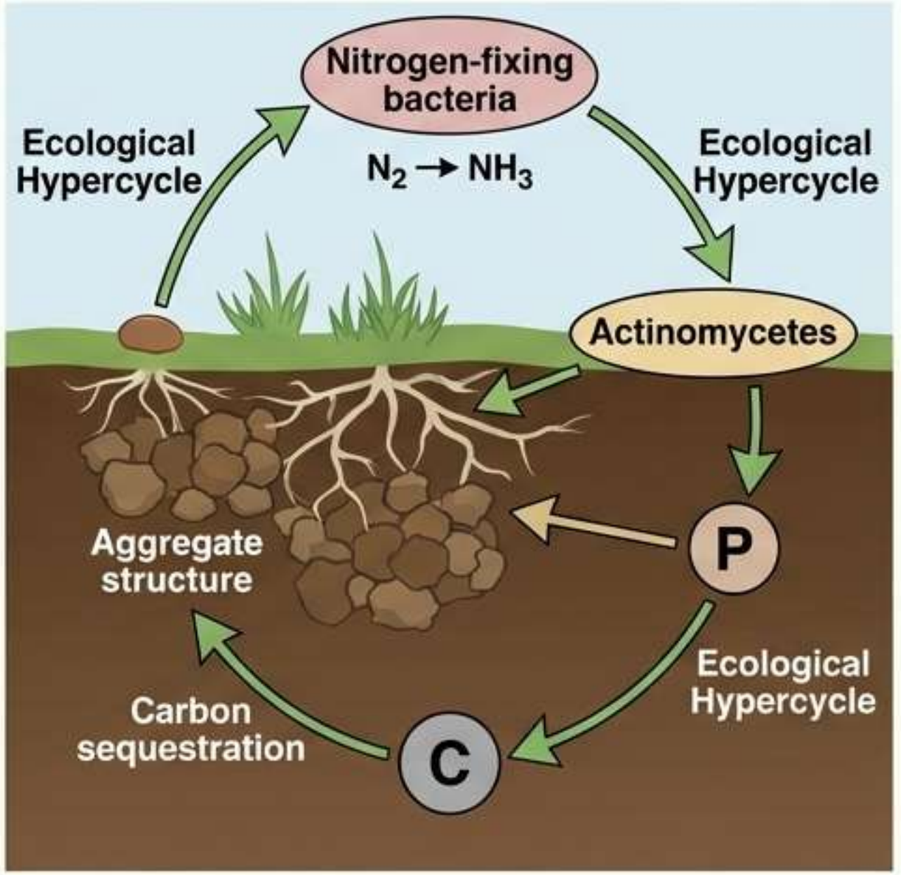


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Layer 1 Soil: Restarting Planetary Metabolism

MBT55 restores the health of all types of soil.

- **Improvement of Physical Properties:** Filamentous fungi and actinomycetes form a hyphal network, binding soil particles. This promotes an 'aggregate structure' with excellent air permeability and water retention.
- **Promotion of Nutrient Circulation:** Nitrogen-fixing bacteria supply atmospheric nitrogen, and actinomycetes convert insoluble phosphate into forms that plants can absorb. This dramatically reduces dependence on chemical fertilizers.
- **Initiation of Ecological Hypercycle**: At the soil level, a self-reinforcing loop is formed where elements such as Carbon (C), Nitrogen (N), and Phosphorus (P) circulate without loss. This signifies the recovery of the life-sustaining functions inherent in the soil.

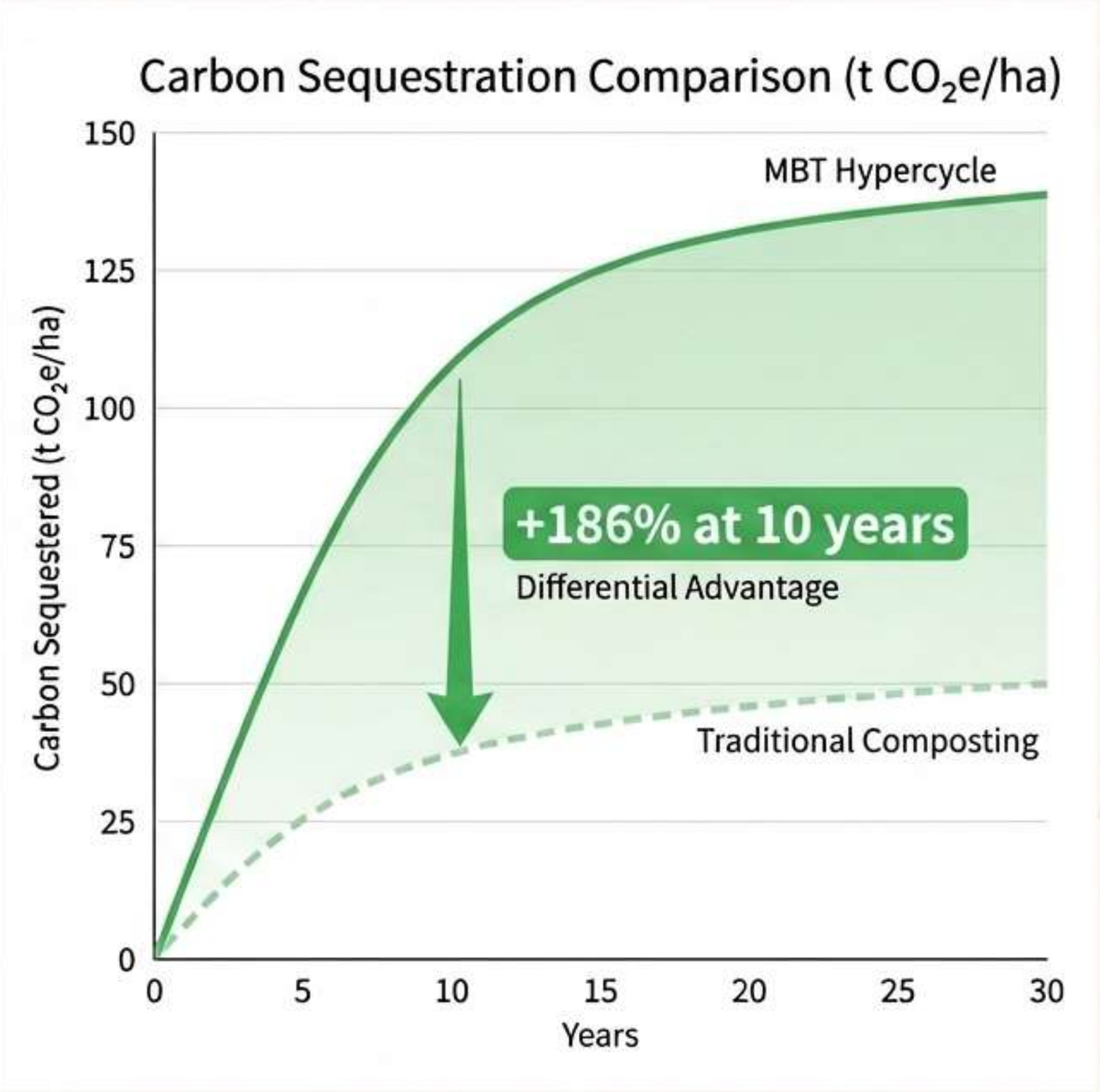


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Quantitative Proof of Carbon Sequestration Capacity: +186% Difference Over 10 Years

The MBT ecological hypercycle creates a decisive difference in long-term carbon storage capacity compared to conventional methods.

- **Mechanism:** MBT55 not only decomposes organic matter but also resynthesizes it into highly stable 'humus' through microbial metabolism. This rate of humic substance formation reaches 2.8 times that of conventional methods.
- **Comparison by Mathematical Model:** According to a carbon sequestration dynamics simulation, the cumulative carbon sequestration over 10 years is 38.2 tCO_2e/ha for conventional composting, while the MBT Hypercycle reaches 109.5 tCO_2e/ha.
- **Warming Resistance:** Even as temperatures rise, the decrease in carbon storage capacity of the MBT system remains at 8%, while conventional methods decrease by 38%. Stability under climate change is outstanding.



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Layer 2 Agriculture: Unmanageable Systemic Crisis

Healthy soil is the foundation for solving the deep-rooted problems facing agriculture.



Case Study 1: Colony Collapse Disorder (CCD):

In 2024-25, 62% of bee colonies in the US disappeared. The cause is a virus transmitted by Varroa mites that are resistant to the acaricide amitraz. Pesticides and malnutrition increase vulnerability, leading to a vicious cycle of ‘virus-mite-nutrition.”

Case Study 2: Coffee Rust:

A fungal disease that reduces yields by up to 70%. Due to climate change, the infection has expanded to higher altitudes, and small-scale farmers are forced to abandon farming due to increased countermeasure costs.



These are not just pests and diseases, but ‘systemic failures’ caused by the collapse of the ecoremic failuree.

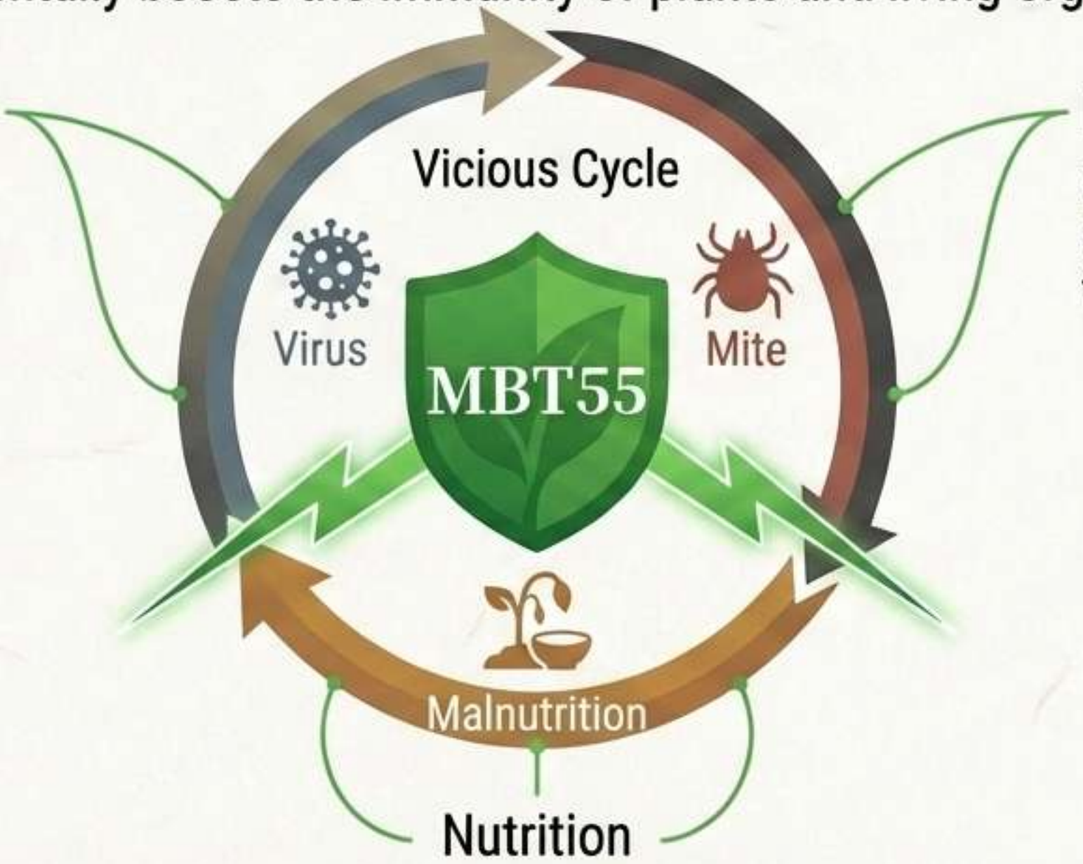
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Multilayer Defense System to Break the Vicious Cycle

MBT55 is not just a fertilizer or pesticide. By restoring the health of the entire soil ecosystem, it fundamentally boosts the immunity of plants and living organisms.

Antiviral & Mite Repellent:

MBT55's metabolites inhibit viral activity and chemically repel mites.



Immune Enhancement:

Healthy nectar plants (like clover) enhance honey bee immunity through nectar.

Nutrition

Fermented pollen paste supplies essential amino acids and improves nutritional vulnerability.

This is a multilayer defense system that breaks the 'Virus-Mite-Nutrition' vicious cycle, offering a sustainable solution distinct from chemical miticides.

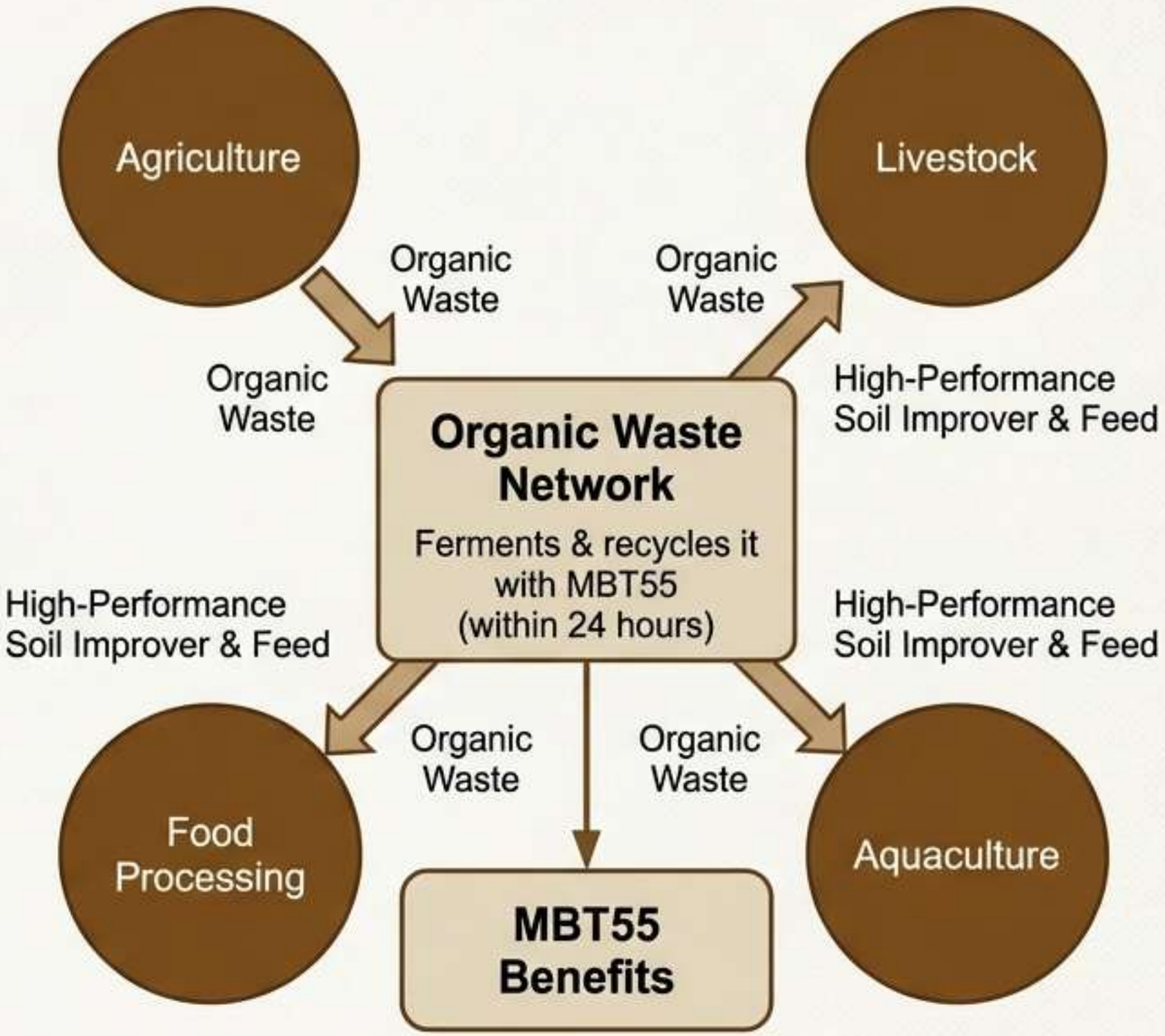
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Level 3 Regional Economy: MBT Sustainable Cycle (MSC)

MBT55 goes beyond individual farms to build a circular model that regenerates the entire regional economy.

MBT Sustainable Cycle (MSC): Organic waste generated from various sectors such as agriculture, livestock, food processing, and aquaculture is collected through a regional "Organic Matter Network."

24-Hour Value Conversion: Through MBT55's high-speed fermentation technology, waste is converted within 24 hours into high-value resources such as high-performance soil improvers and feed. This reduces waste disposal costs and creates new resources within the region.
Waste is no longer a cost, but an input that enriches the regional economy.



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Layer 4 Global Infrastructure: AGRIX – OS for Natural Capital

To manage and optimize at scale the countless ecological hypercycles built at the regional level, a new type of intelligence is needed: an "OS for Natural Capital."

AgriWare™ (Agricultural OS)

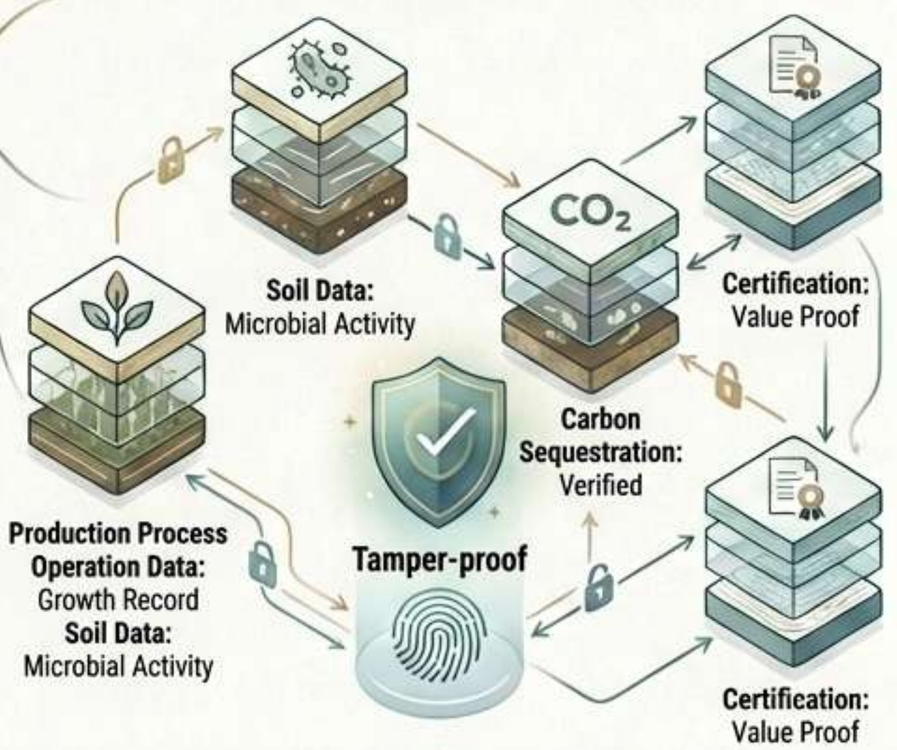
Utilizing "dynamic phenotyping" technology, it diagnoses the "response behavior and temporal trends" of the soil-microbe-crop system, rather than just its "state." A cockpit that captures the ecosystem's "dynamic vitals" and enables proactive, optimal intervention.



AGRIX Platform: An operating system that integratedly manages microbes, soil, crops, climate, and economic activities.

SafelyChain™ (Post-Traceability Infrastructure)

A 'digital notary' that records the entire production process in a tamper-proof manner. It proves not just the origin, but the "value creation process" itself, such as carbon sequestration and nutrient addition, building a foundation of trust.

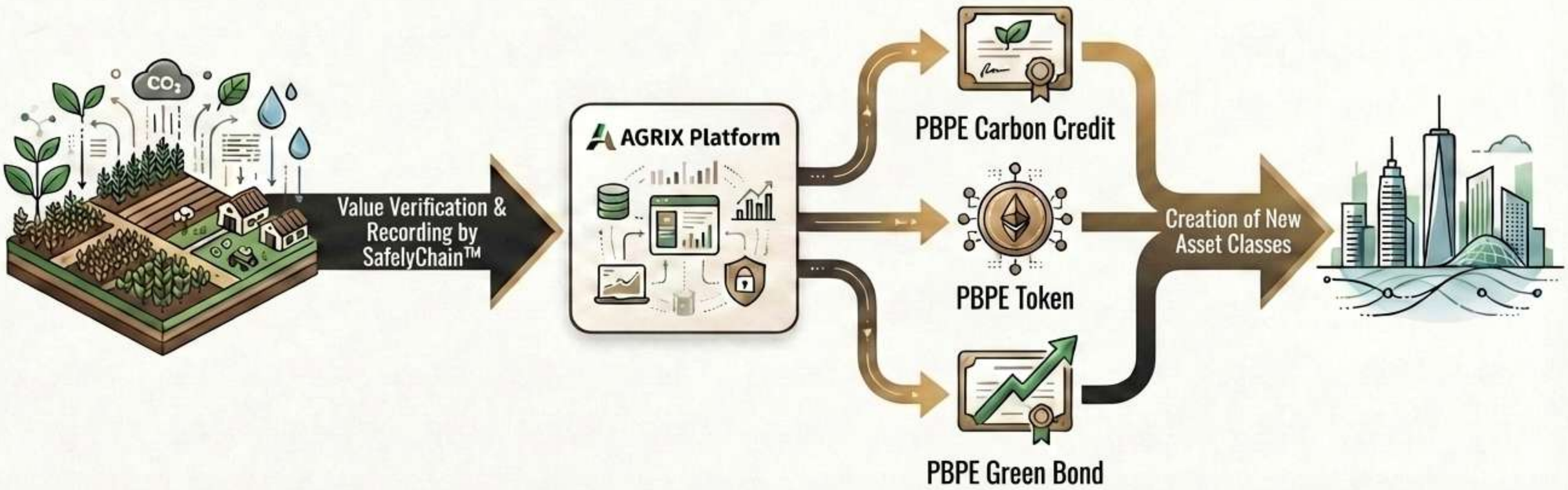


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Final Layer New Economy: Financialization of Regenerative Value

AGRIX directly connects agricultural practices and global environmental regeneration with global financial markets.

- "Agricultural Product Trading = Emission Rights Trading": Environmental values such as "carbon sequestration" and "food loss reduction" verified and recorded by SafelyChain™ become new assets attached to the agricultural products themselves.
- "Creation of a New Asset Class": This verified data becomes the raw material for new financial products.



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Currency Recording Value Creation: Proof of Regeneration

PBPE financial products are designed as an infrastructure to financially reward "**good production activities**," not speculation.

Essence of PBPE Token: If Bitcoin is "Proof of Work," PBPE is "Proof of Regeneration."

Source of Value: Tokens are issued based on the following "Earth-positive outcomes."

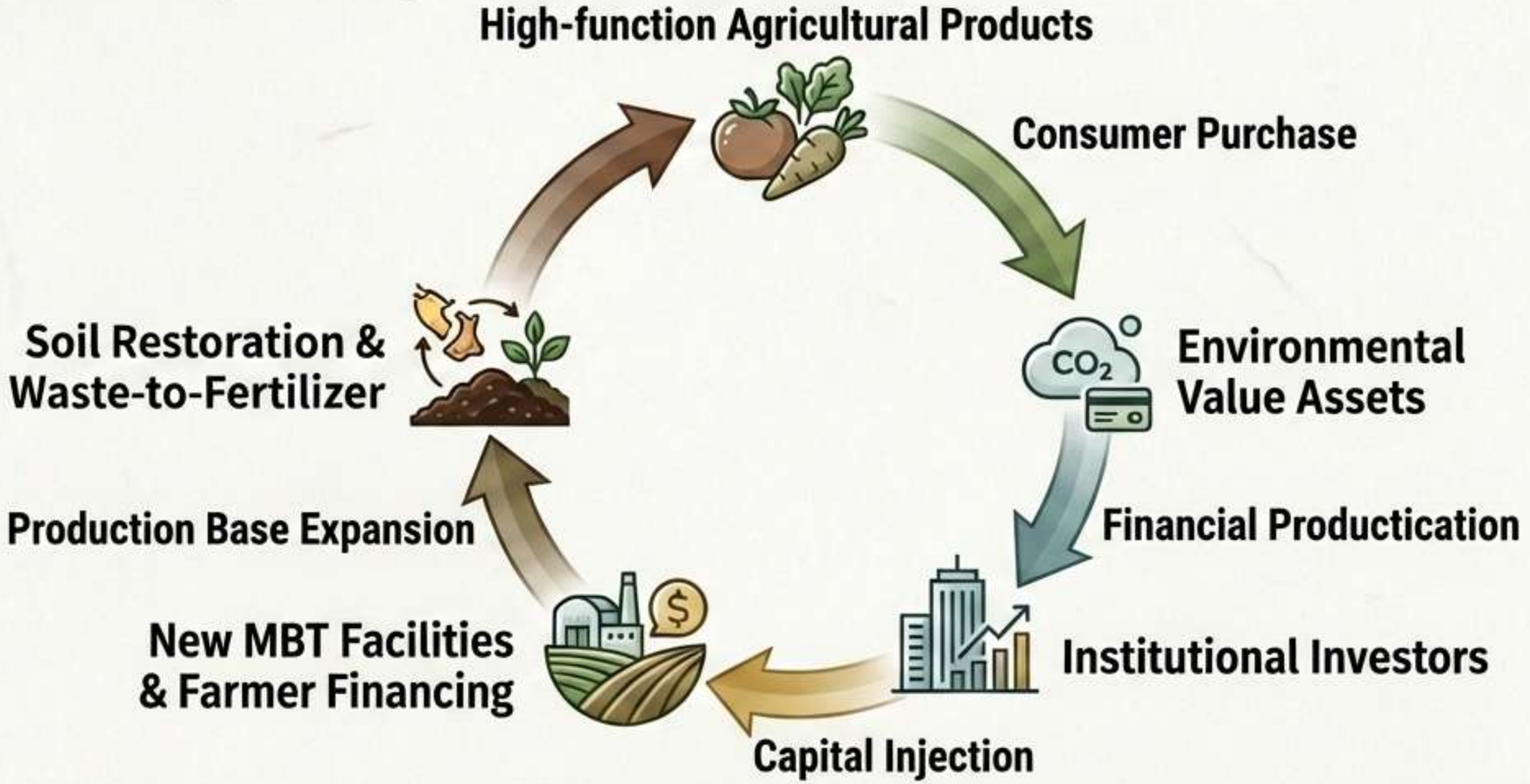


Issuance Model: [$T_{mint} = a \cdot \Delta C + b \cdot \Delta Y + c \cdot \Delta L$]
The issuance amount is determined based on outcomes verified by AGRIX, as shown above.
This is a "currency tied to work," where production activities themselves create value.

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Ultimate Flywheel: MBT Multi-layered Value Cycle (MMVC)

By integrating the engine (MBT55) and the OS (AGRIX), the ultimate flywheel is completed, where economic activity itself regenerates the Earth.



In this loop, farmers becore: “Food, Environment, and Health,’ and simulaniously create three operators and consumer choices and financial investments directly lead to planetary regeneration.

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We propose not just “carbon reduction technology,” but a “civilization that circulates life.”

- **Nature as Infrastructure:** Healthy soil is the most critical social infrastructure supporting disaster prevention, water, and health.
- **Value by Circulation:** Waste does not exist. Everything is a resource to generate the next value.
- **Co-evolutionary Governance:** Human, microorganism, and AI collaborate to design a society where economy and ecosystem co-evolve.

This is a blueprint for a future where economic activities are perfectly integrated with the restoration of natural ecosystems.

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